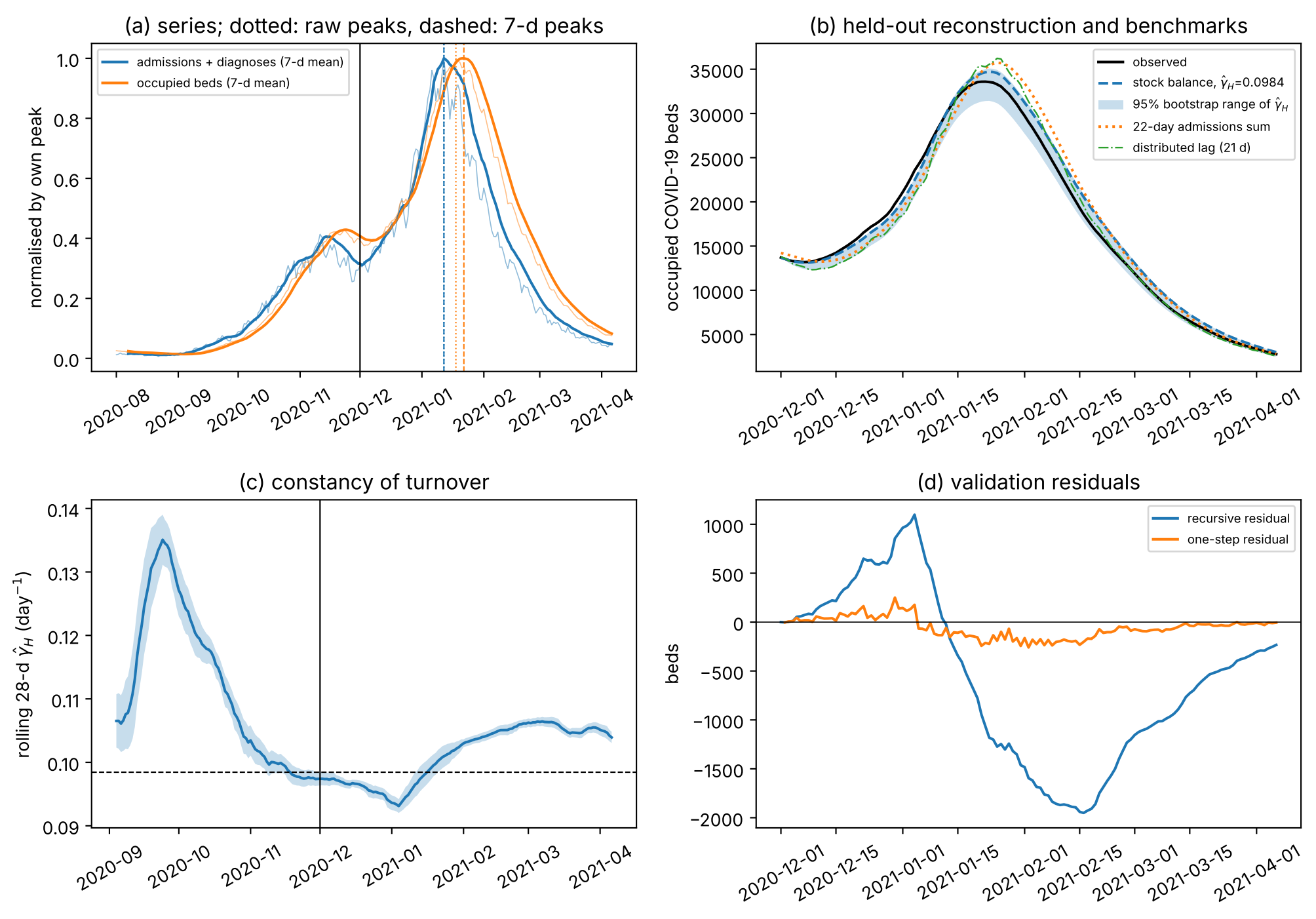
(a) series; dotted: raw peaks, dashed: 7-d peaks
admissions + diagnoses (7-d mean)
occupied beds (7-d mean)
normalised by own peak
1.0
0.8
0.6
0.4
0.2
0.0
2020-08
2020-09
2020-10
2020-11
2020-12
2021-01
2021-02
2021-03
2021-04
(b) held-out reconstruction and benchmarks
observed
stock balance, $\hat{\gamma}_H$=0.0984
95% bootstrap range of $\hat{\gamma}_H$
22-day admissions sum
distributed lag (21 d)
occupied COVID-19 beds
35000
30000
25000
20000
15000
10000
5000
2020-12-01
2020-12-15
2021-01-01
2021-01-15
2021-02-01
2021-02-15
2021-03-01
2021-03-15
2021-04-01
(c) constancy of turnover
rolling 28-d $\hat{\gamma}_H$ (day^{-1})
0.14
0.13
0.12
0.11
0.10
0.09
2020-09
2020-10
2020-11
2020-12
2021-01
2021-02
2021-03
2021-04
(d) validation residuals
recursive residual
one-step residual
beds
1000
500
0
−500
−1000
−1500
−2000
2020-12-01
2020-12-15
2021-01-01
2021-01-15
2021-02-01
2021-02-15
2021-03-01
2021-03-15
2021-04-01